KENTUCKY COMMUNITY & TECHNICAL COLLEGE SYSTEM

ESSENTIAL SKILL 8 · FROM YOUR COLLEAGUES

Valley Creek Environmental Hazard Role-Play

GLY 110 · Environmental Geology

Students take the roles of community stakeholders, from the county emergency manager to a local business owner, after flooding, landslide risk, and possible groundwater contamination hit a fictional Kentucky town. Each team negotiates a shared set of recommendations for county leaders.

Contributed by **Sherri Basil** and adapted with permission for *Designing Collaboration That Works*, the KCTCS faculty guide to Essential Skill 8. Student-facing sections are written to copy into Blackboard. Draft for review, in-house resource.

COURSE
GLY 110 Environmental Geology

LENGTH
One class session in person, or about a week online

TEAM SIZE
4–6 students, one role each

FORMAT
In person or online

FINAL PRODUCT
Team recommendation plan, as a report or a briefing

Closest level in the guide: **Level 3 · Coordinated**

1 · Quick	2 · Structured	**3 · Coordinated**	4 · Sustained

In this packet

1. Faculty Overview FOR FACULTY
2. Suggested Essential Skill 8 Statement FOR FACULTY
3. Instructor Preparation and Roles FOR FACULTY
4. Student Instructions FOR STUDENTS

Faculty Overview

FOR FACULTY

GLY 110 Environmental Geology Role-Play Activity

Course Alignment

This role-play activity aligns with the following GLY 110 learning outcomes:

- Describe geologic processes that affect conditions on Earth, including natural hazards such as flooding, landslides, and groundwater issues.
- Explain how Earth resources are used to meet human needs.
- Demonstrate understanding of how human activities interact with Earth systems and create environmental consequences.

Faculty Overview

In this scenario, students assume the roles of community stakeholders responding to an environmental crisis in a fictional Kentucky community. Students must analyze geologic hazards, evaluate environmental impacts, consider resource management issues, and work collaboratively to develop recommendations for local officials.

The activity is designed to move beyond memorization and encourage students to apply geology concepts to realistic situations that require critical thinking, problem solving, and communication.

Suggested Format

Face-to-Face Course

- Divide students into groups of 4-6.
- Assign one stakeholder role to each student.
- Allow 20-30 minutes for group discussion.
- Have groups present a 5-minute emergency briefing.

Online or Blackboard Course

- Create discussion groups.
- Assign roles within each group.
- Require students to post an initial response from their assigned perspective.
- Students collaborate to create a final group recommendation.
- Groups submit a written report or recorded presentation.

Faculty Note: Essential Skill 8 Alignment

This activity is designed to develop students' ability to collaborate with individuals who have different priorities, perspectives, and expertise. Students must communicate effectively, consider multiple viewpoints, negotiate solutions, and contribute to a shared decision-making process.

Rather than working independently, students must engage in meaningful dialogue to create a consensus-based response to a complex environmental problem.

Suggested Essential Skill 8 Statement

FOR FACULTY

Use this statement in your syllabus or in an announcement to students.

This activity supports Essential Skill 8: Collaborate and Work in Teams by requiring students to work with others, evaluate diverse perspectives, negotiate solutions, and create a shared recommendation to address a complex environmental geology problem. Through structured collaboration, students practice the communication, teamwork, and problem-solving skills necessary for success in both academic and workplace environments.

Instructor Preparation and Roles

FOR FACULTY

Before the activity:

- Review major concepts related to flooding, landslides, water resources, mining impacts, and environmental hazards.
- Prepare role cards or post role descriptions in Blackboard.
- Determine group assignments.
- Decide whether students will submit a presentation, report, discussion post, or infographic.

Student Roles

Assign students to groups of 4-6. Each student receives one role.

County Emergency Manager

- Focus on public safety.
- Prioritize evacuations, emergency response planning, and community preparedness.
- Concerned about protecting lives during flooding and landslides.

Geologist

- Explains the Earth processes causing the hazards.
- Identifies areas most at risk.
- Recommends monitoring and mitigation strategies.

Environmental Scientist

- Investigates potential groundwater and soil contamination.
- Evaluates environmental impacts.

- Suggests cleanup and protection measures.

Energy and Resource Specialist

- Examines how natural resources are being used.
- Discusses how mining and land development contribute to current conditions.
- Recommends sustainable resource management practices.

Local Business Owner

- Represents economic concerns.
- Wants solutions that protect jobs and property.
- Advocates for practical and affordable actions.

Community Resident

- Represents affected citizens.
- Focuses on health, safety, and quality of life.
- Raises concerns about drinking water and property damage.

Student Instructions

FOR STUDENTS

Scenario

You are a member of the **Valley Creek Environmental Hazard Response Team**, a group of experts and community representatives assembled to address environmental challenges affecting the community. Recent heavy rainfall has caused flooding, increased landslide risks, and raised concerns about groundwater contamination near an abandoned mining site.

Working with your team, you will analyze the situation and develop recommendations for local officials.

Your Task

As a team, create a recommendation plan for Valley Creek. Your final submission should address all four parts of the investigation.

Part 1: Identify the Geological Hazards

As a team, investigate the geological hazards affecting Valley Creek.

Discuss the following:

- What geological processes are occurring?
- Why are flooding and landslides happening?
- Which areas of the community face the greatest risk?
- How do these hazards affect people, property, and the environment?

Deliverable: Write a summary explaining the hazards and the geological processes responsible for them.

Part 2: Analyze Human Impacts

Examine how human activities may have contributed to the environmental challenges.

Consider the effects of:

- Mining
- Construction
- Deforestation
- Urban development
- Changes in land use

Discuss:

- How these activities may have increased the risks of flooding, landslides, or contamination.
- How human actions can influence natural geological processes.

- What environmental consequences may result from these activities.

Deliverable: Explain the relationship between human activity and the current environmental problems.

Part 3: Evaluate Earth Resources

Identify the Earth resources that may be affected by the environmental hazards.

Examples may include:

- Groundwater
- Surface water
- Soil
- Minerals
- Forest resources
- Agricultural land

Discuss:

- How these resources support the community.
- Why these resources are important for daily life and economic activity.
- What long-term impacts may occur if the hazards are not addressed.

Deliverable: Provide an analysis of the affected resources and explain why they should be protected.

Part 4: Recommend Solutions

Develop at least **three recommendations** for county leaders.

Possible recommendations include:

- Hazard monitoring programs
- Improved drainage systems
- Water testing initiatives
- Land-use regulations
- Emergency preparedness plans
- Environmental cleanup or remediation projects

For each recommendation:

1. Describe the proposed action.
2. Explain how it addresses the geological hazard.
3. Discuss potential benefits to the community.

Deliverable: Present at least three evidence-based recommendations supported by your team's analysis.

Final Submission

Your team's final recommendation should include:

Introduction

- Brief overview of the Valley Creek situation.

Part 1: Geological Hazards

- Description of hazards and affected areas.

Part 2: Human Impacts

- Analysis of how human activities contributed to the problem.

Part 3: Earth Resources

- Evaluation of impacted resources and community effects.

Part 4: Recommendations

- At least three solutions with supporting reasoning.

Team Reflection

Conclude with a brief reflection describing how your team collaborated to reach a final recommendation and how different perspectives contributed to your solution.